SpaceX Falcon 9 First Stage Landing Prediction

GitHub Repository:

https://github.com/Satish-KumarRam/IBM-Data-Science-Capstone-Project

IBM Data Science Professional Certificate Capstone Project

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OUTLINE

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

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EXECUTIVE SUMMARY

- **Objective**
- Predict Falcon 9 first-stage landing success using historical launch data.
- **Methodology Summary**
    - Data Collection through **SpaceX REST API**
    - Data Collection with **Web Scraping (Wikipedia)**
    - Data Wrangling
    - Exploratory Data Analysis (**Visualization & SQL**)
    - Interactive Visual Analytics (**Folium & Plotly Dash**)
    - Predictive Analysis using Machine Learning
- **Key Results**
    - Landing success increased over time.
    - Payload mass, orbit, and launch site significantly affected landing success.
    - The best machine learning model achieved the highest prediction accuracy.

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INTRODUCTION

- **Project Background:** SpaceX reduces launch costs by successfully recovering and reusing Falcon 9 first-stage boosters, making launches more efficient and cost-effective.
- **Problem Statement:** Predict whether the Falcon 9 first-stage landing will be successful before launch using historical mission data and machine learning techniques.
- **Project Objective:** Build, train, and evaluate machine learning classification models to predict Falcon 9 first-stage landing success using historical launch data.
- **Data Sources:**
  - **SpaceX REST API:** Launch information, payload details, booster version, launch site, and mission outcomes.
  - **Wikipedia (Web Scraping):** Additional Falcon 9 launch records used to validate and enrich the dataset.

METHODOLOGY

- **Data Collection:**
  - Collected Falcon 9 launch data using the **SpaceX REST API**.
  - Collected additional launch records through **Wikipedia Web Scraping**.
- **Data Wrangling:**
  - Cleaned missing values and standardized the dataset.
  - Merged API and Wikipedia datasets.
- **Exploratory Data Analysis (EDA):**
  - Created visualizations using Matplotlib and Seaborn.
  - Analyzed relationships between mission features and landing success.
- **SQL Analysis:**
  - Queried the database to analyze launch statistics.
  - Retrieved insights using SQL queries.
- **Interactive Visualization:**
  - Built interactive **Folium** maps.
  - Developed a **Plotly Dash** dashboard.
- **Predictive Analysis:**
  - Built and evaluated multiple classification models.
  - Selected the best-performing model.

RESULTS

- **Landing Success Trend:** Landing success improved steadily over time.
- **Key Factors:** Payload mass, launch site, orbit type, flight number, and booster version influenced landing success.
- **EDA, SQL & Visualization:** SQL queries, Folium maps, and the Plotly Dash dashboard provided interactive insights.
- **Predictive Analysis:** The best classification model achieved the highest prediction accuracy.
- **Overall Outcome:** Historical launch data can effectively predict Falcon 9 first-stage landing success and support mission planning.

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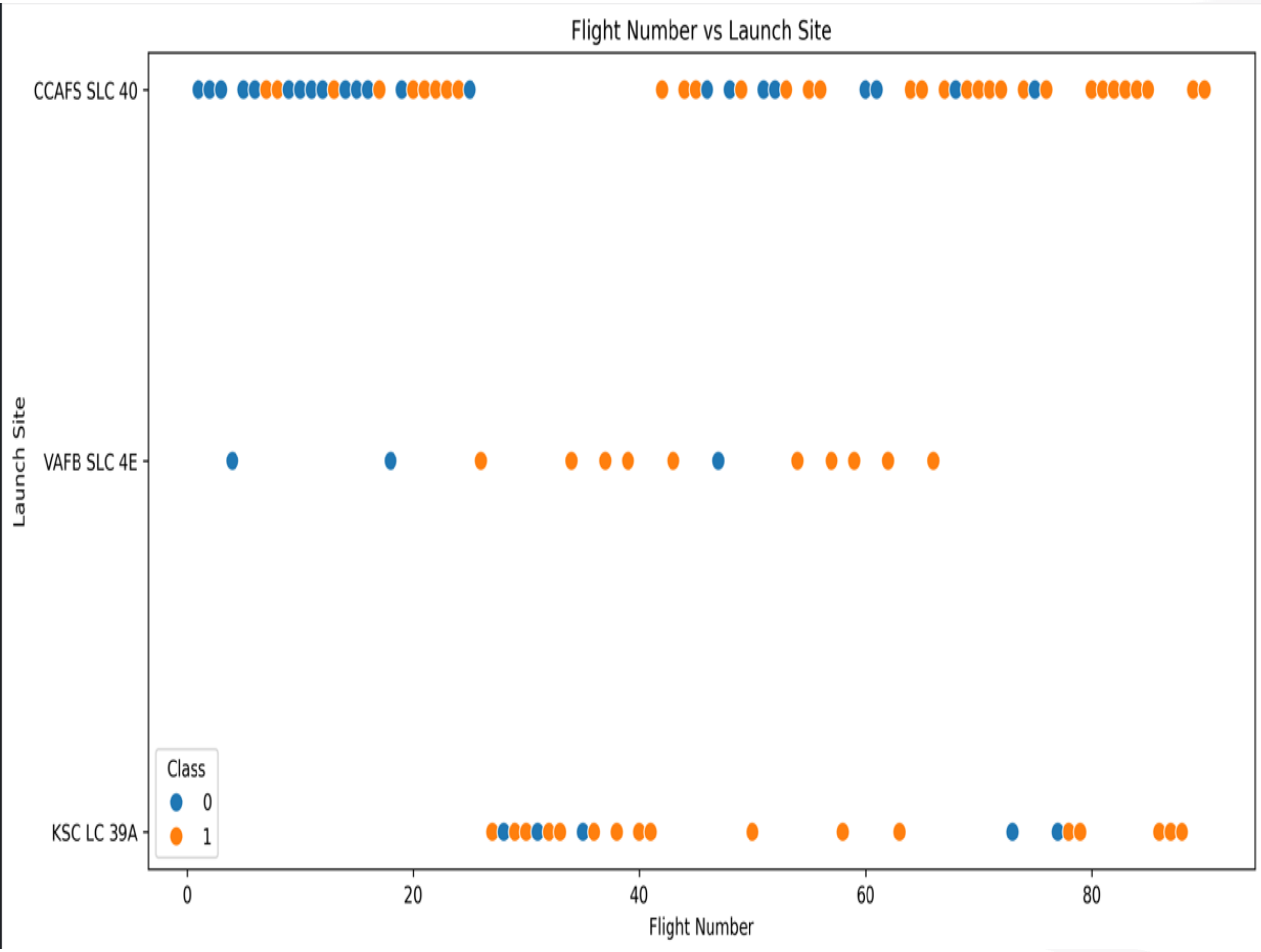
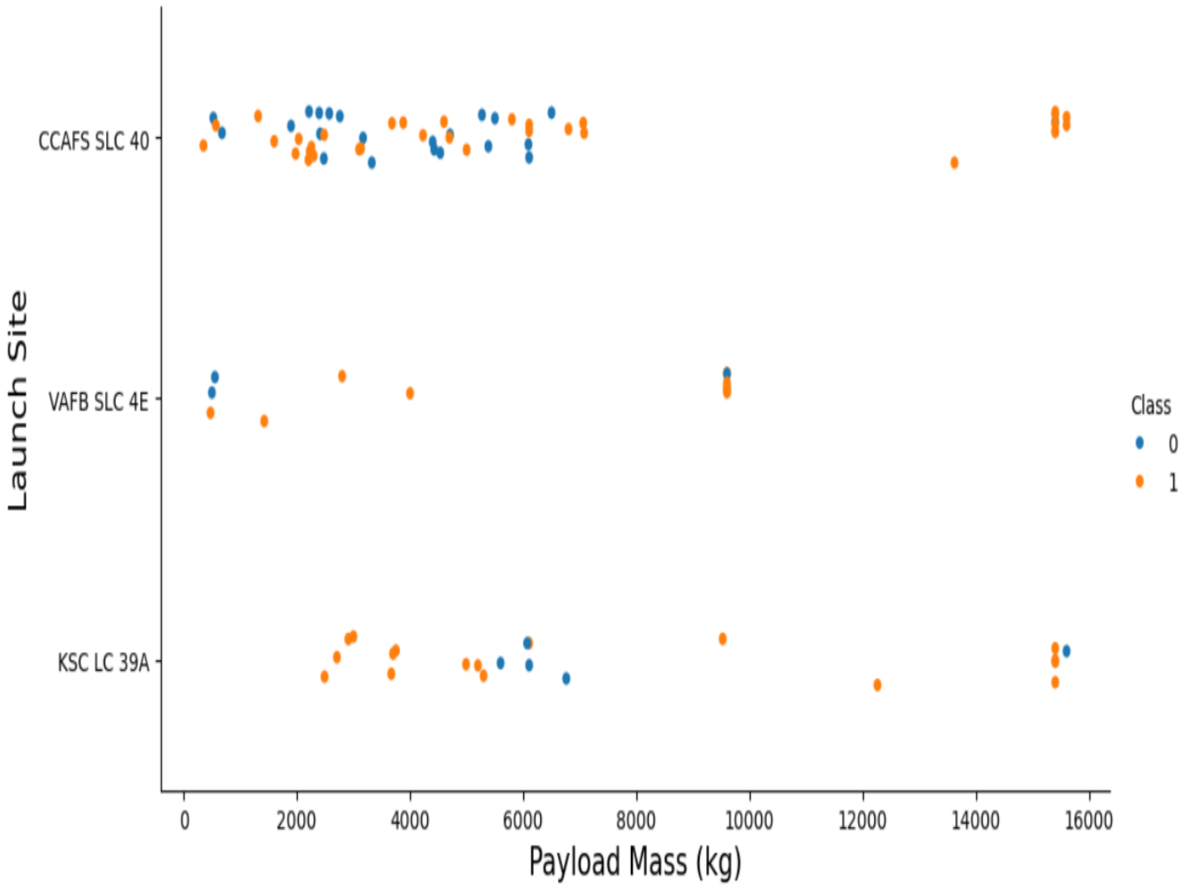
Flight Number vs. Launch Site



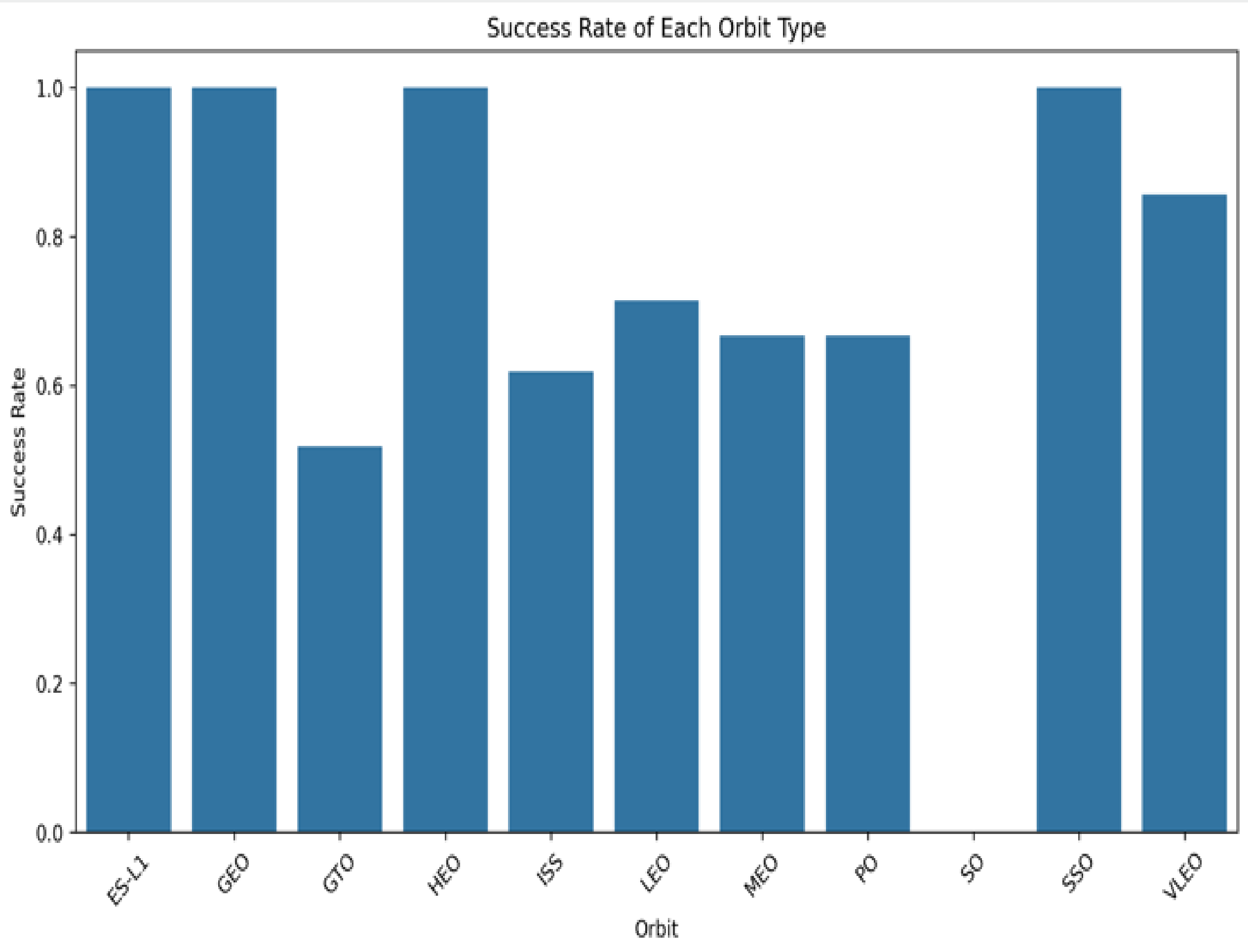
- **Key Findings**
- Early Falcon 9 missions had lower landing success, while later missions showed higher success rates.
- Launch success improved as the **flight number** increased, indicating continuous improvements in booster recovery and operational experience.
- **CCAFS SLC 40** recorded the highest number of Falcon 9 launches, making it the most frequently used launch site.
- **KSC LC 39A** and **VAFB SLC 4E** also demonstrated successful landings, although with fewer launches than CCAFS SLC 40.
- Overall, both launch site and flight number influenced Falcon 9 first-stage landing success.

Payload Mass vs. Launch Site



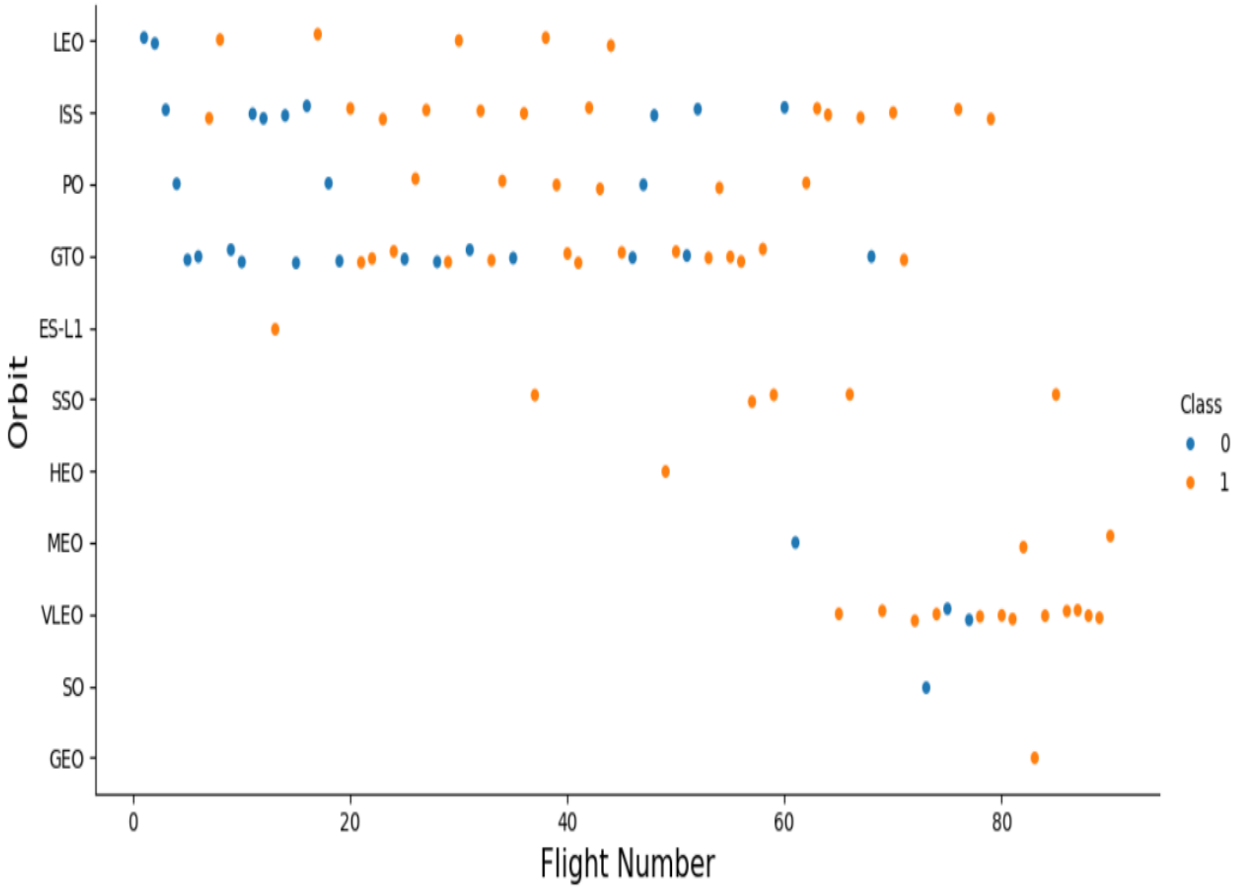
- **Key Findings**
- **CCAFS SLC 40** handled the widest range of payload masses and was the most frequently used launch site.
- **KSC LC 39A** primarily handled medium- to high-payload missions and achieved several successful landings.
- **VAFB SLC 4E** conducted fewer launches but demonstrated successful missions across different payload categories.
- Both **successful** and **unsuccessful** landings occurred at low, medium, and high payload masses, indicating that **payload mass alone was not the primary factor affecting landing success**.
- Overall, launch site and payload mass together influenced Falcon 9 landing success.

Success Rate by Orbit



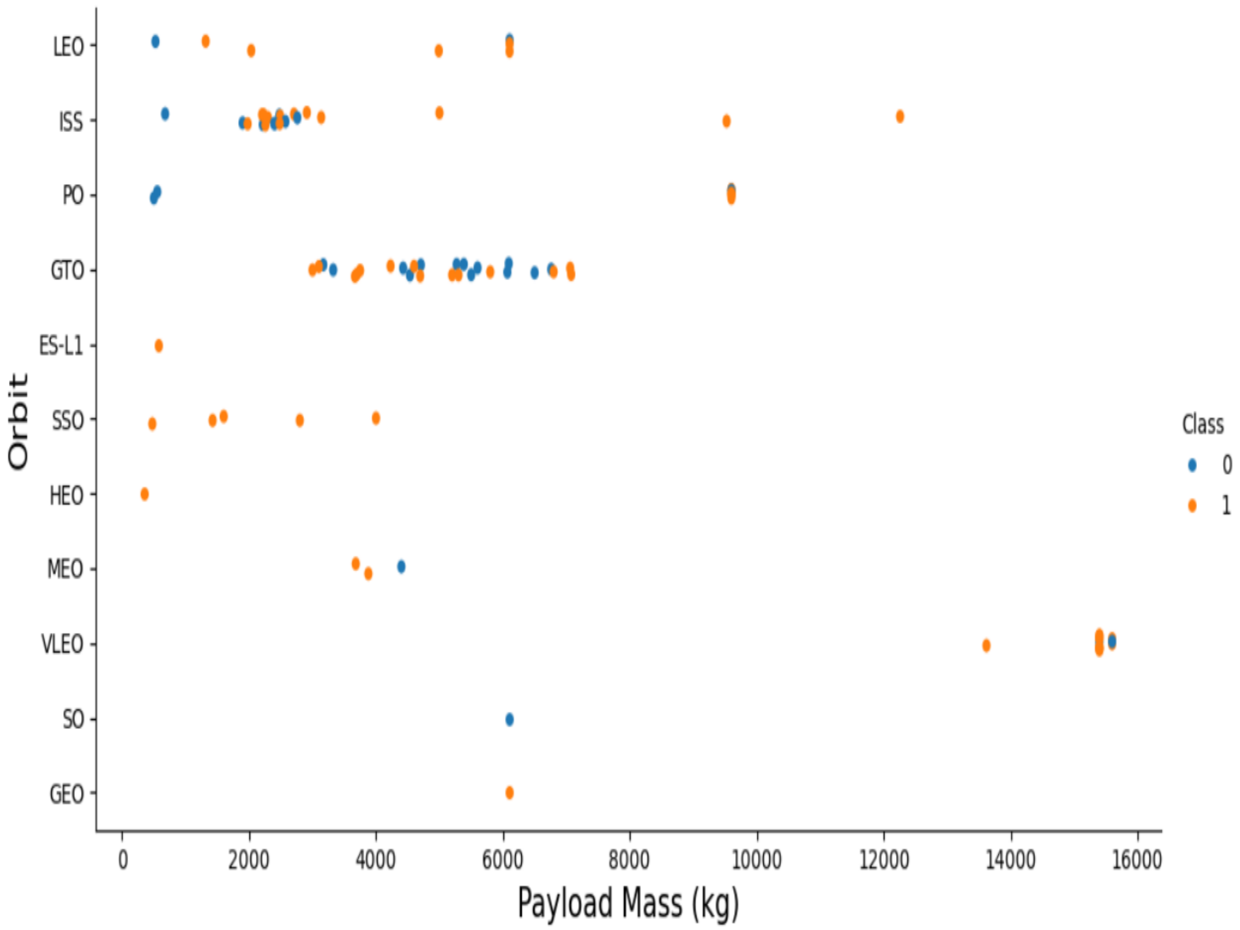
- **Key Findings**
- **ES-L1, GEO, HEO,** and **SSO** achieved a **100% landing success rate**, indicating highly reliable mission outcomes for these orbit types.
- **VLEO** also demonstrated a high landing success rate of approximately **86%**, showing strong booster recovery performance.
- LEO, MEO, ISS, and PO achieved moderate landing success rates, indicating variable mission outcomes across these orbit types.
- **GTO** recorded a comparatively lower landing success rate, while **SO** showed **no successful landings** in the available dataset.
- Overall, orbit type significantly influenced Falcon 9 landing success, although mission characteristics and operational factors also played important roles.

Flight Number vs. Orbit



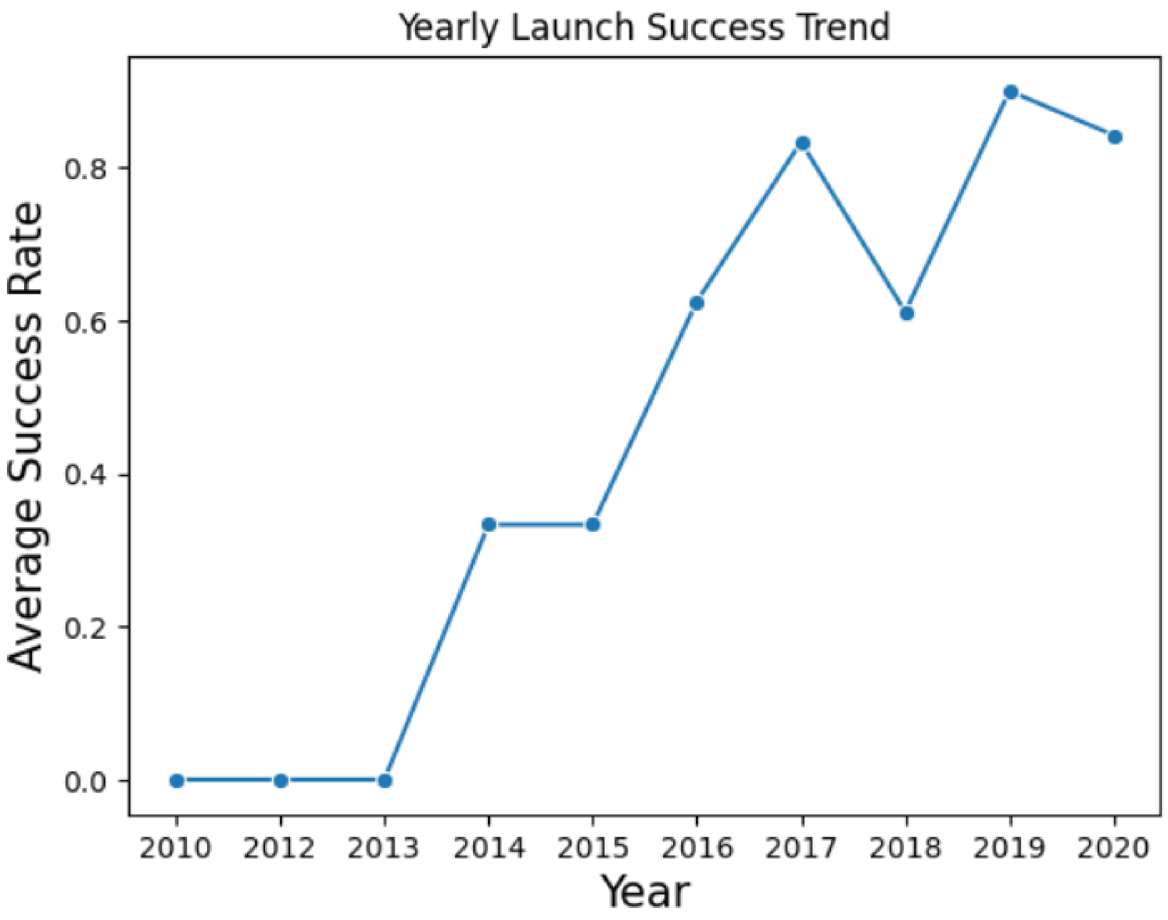
- **Key Findings**
- LEO, ISS, and VLEO missions generally achieved high landing success rates.
- GTO missions showed mixed landing outcomes, with both successful and failed landings.
- Landing success generally improved with increasing flight numbers, especially in later missions.
- Orbit type influenced landing success, while payload mass and launch site also played important roles.

Payload Mass vs. Orbit



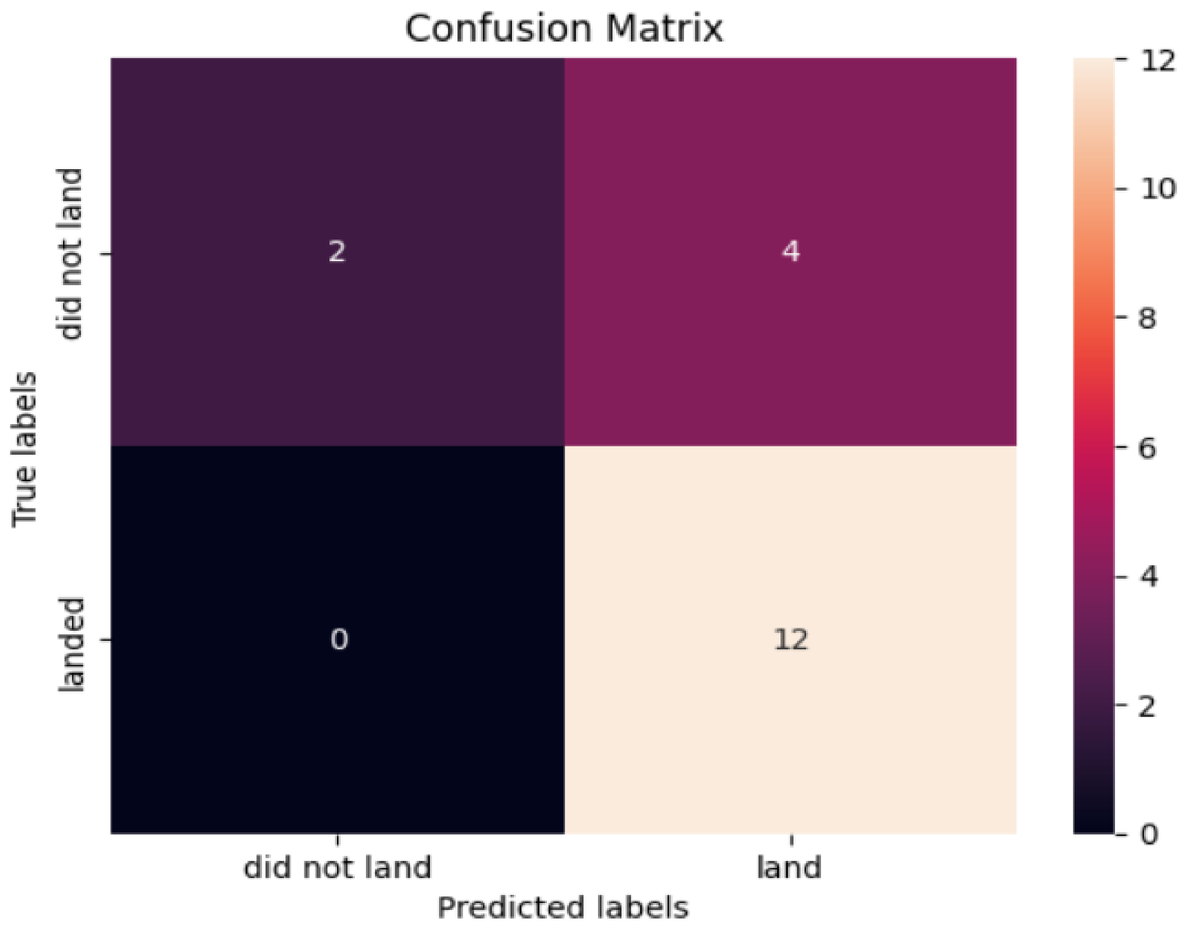
- **Key Findings**
- GTO missions covered a wide range of payload masses, with both successful and unsuccessful landings observed.
- VLEO missions were associated with the highest payload masses.
- LEO and ISS missions generally carried low to medium payload masses.
- Payload mass alone did not determine Falcon 9 landing success; orbit type and mission characteristics also influenced landing outcomes.

Yearly Launch Success Trend



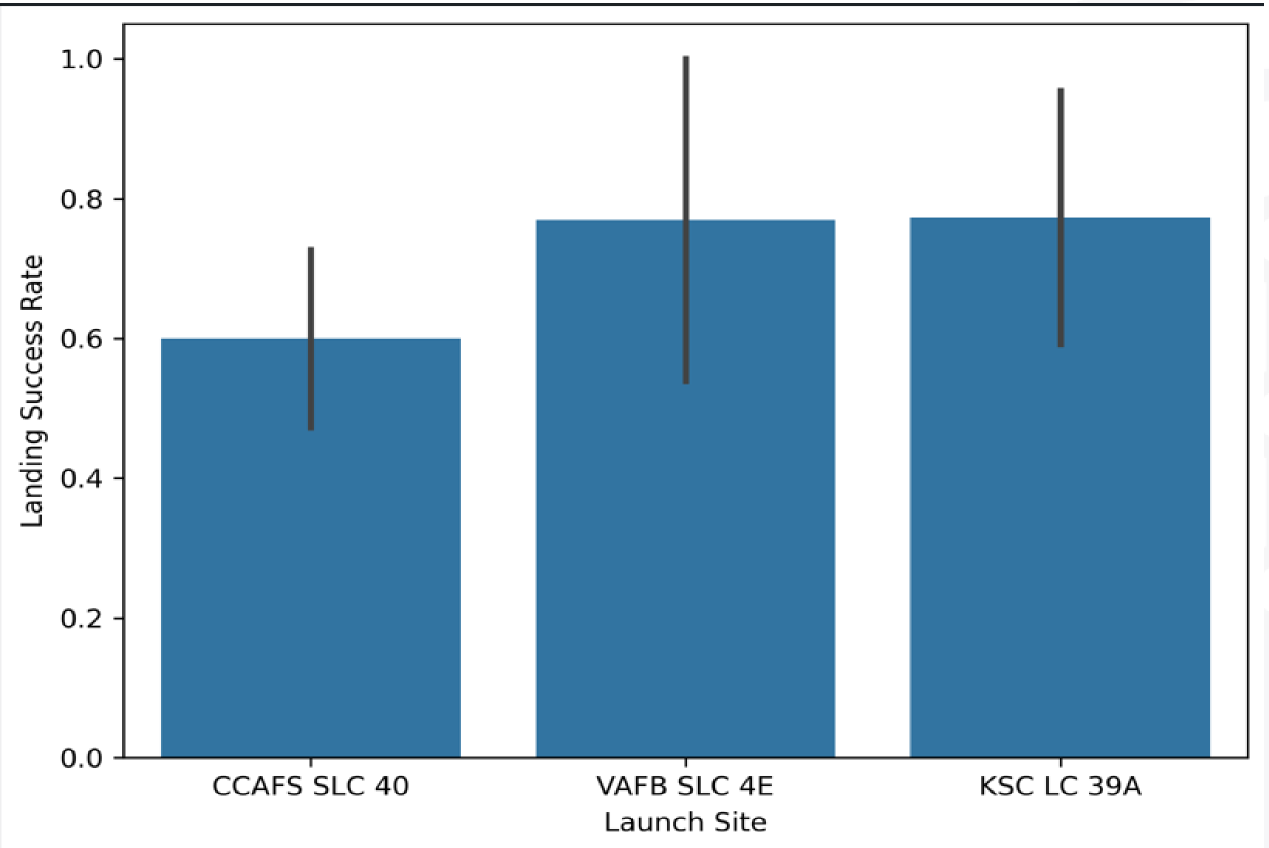
- **Key Findings**
- Overall launch success rates improved over the years despite minor fluctuations.
- Success rates were very low during the early years (2010–2013).
- From 2016 onward, average launch success rates increased significantly.
- By 2019–2020, SpaceX consistently achieved high launch success rates, demonstrating improved booster recovery and mission reliability.

Model Evaluation – Confusion Matrix



- **Key Findings**
- The model correctly predicted **12 successful landings (True Positives)**.
- **2 unsuccessful landings** were correctly classified (True Negatives).
- **4 unsuccessful landings** were incorrectly predicted as successful (False Positives).
- The model demonstrated strong performance in identifying successful landings, with no false negatives, although some unsuccessful missions were incorrectly classified as successful.

Landing Success by Launch Site



- **Key Findings**
- **KSC LC 39A** achieved the **highest average landing success rate** among all launch sites.
- **VAFB SLC 4E** also recorded a **high landing success rate**, indicating strong mission performance.
- **CCAFS SLC 40** showed the **lowest average landing success rate** in the dataset.
- Launch site significantly influenced Falcon 9 first-stage landing success.

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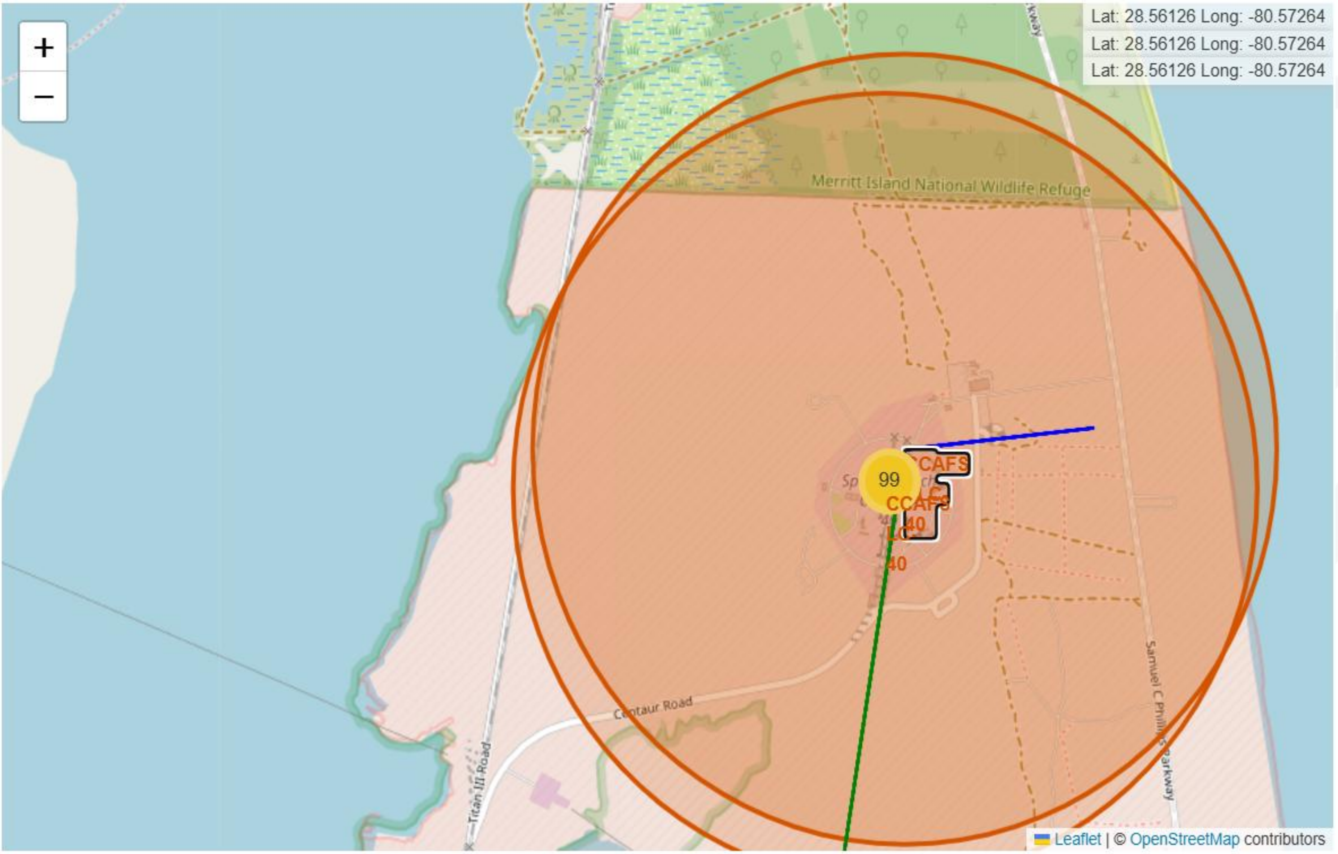
Launch Site Proximity Analysis (CCAFS)



- **Key Findings:**
- CCAFS launch site is located close to the coastline.
- The coastal location supports safe launch trajectories over the ocean.
- The site is positioned away from major populated areas to improve safety.
- The proximity map highlights CCAFS's strategic coastal location for safe launch operations.

SPACEX FALCON 9 – FINDINGS & IMPLICATIONS

Findings

- Launch success rate improved over time.
- Launch site, payload mass, flight number, and orbit type significantly influenced landing success.
- The best-performing classification model demonstrated strong predictive performance.

Implications

- Machine learning can effectively predict Falcon 9 landing success.
- The predictive model can support mission planning and reduce launch costs.
- Data-driven insights help improve future SpaceX launch decisions.

RESULTS – VISUALIZATIONS & DASHBOARD

EDA Visualizations

- Flight Number vs. Launch Site
- Payload Mass vs. Launch Site
- Success Rate by Orbit
- Flight Number vs. Orbit
- Payload Mass vs. Orbit
- Yearly Launch Success Trend
- Landing Success by Launch Site

Interactive Dashboard

- SQL Query Results
- Folium Launch Site Map
- Plotly Dash Dashboard
- Confusion Matrix (Model Evaluation)

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Conclusion

- **Key Takeaways**
- The analysis identified several factors that influence **SpaceX Falcon 9 landing success**.
- **Flight Number** showed a positive relationship with landing success, indicating improved performance over time.
- Payload mass, orbit type, and launch site significantly influenced landing success.

- **Overall Findings**
- The yearly trend demonstrated a significant improvement in landing success rates. By 2019, launch success rates had reached their highest levels and remained consistently high thereafter.
- Overall, the analysis shows that historical launch data can effectively predict Falcon 9 landing success and support data-driven decision making.

OVERALL FINDINGS & IMPLICATIONS

Findings

- Launch success rate improved significantly over the years.
- Flight number, launch site, orbit type, and payload mass were identified as key factors influencing landing success.
- Machine learning models successfully predicted Falcon 9 landing outcomes with good accuracy.

Implications

- Predictive models can support mission planning and decision-making.
- Understanding launch factors helps improve mission reliability and reduce operational costs.
- Data-driven insights support better decision-making for future Falcon 9 missions.

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DISCUSSION

- **Key Discussion Points**
- Launch success was influenced by payload mass, launch site, flight number, and orbit type.
- EDA identified important trends in the SpaceX launch dataset.
- Plotly Dash and Folium visualizations improved interactive data exploration.
- The best classification model achieved good predictive performance with high prediction accuracy.
- Predictions are based on historical data and may not fully capture future mission conditions.
- Future work can include weather data, real-time launch conditions, and more advanced machine learning models.

CONCLUSION

- Launch success increased over time.
- Payload mass, launch site, and orbit type influenced landing success.
- The best classification model achieved good predictive performance with high prediction accuracy.
- The interactive dashboard and machine learning model support analysis and prediction of Falcon 9 landing success.

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APPENDIX

- **Additional Analysis**
- SQL query outputs used for exploratory analysis.
- Interactive Plotly Dash dashboard for exploratory data analysis.
- Folium map showing launch site proximity.
- Additional EDA charts supporting model development.

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